Do early pension withdrawals reduce labour supply?
Evidence from Australia's pandemic response

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Abstract

We investigate the impact of the early release of Australian pension savings during the COVID-19 pandemic on short- and medium-term labour supply behaviour. Given the endogenous nature of this policy, we also apply an instrumental variable strategy to isolate the causal effect of the policy on employment and weekly hours worked. Our findings show the pension release scheme led to a sizeable and negative extensive margin response to labour supply in 2020, before dissipating in 2021. This points to the reduction in labour supply caused by the policy being a brief shock rather than a persistent effect. We find evidence the effect was primarily concentrated among female withdrawers and those who withdrew the full amount allowable under the policy. Our findings improve our understanding of the labour market impacts of early pension release schemes in a crisis context, and can also inform the design of pension systems more broadly.

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1 Introduction and literature

As part of its economic response to the COVID-19 pandemic, the Australian Government introduced the Early Release of Superannuation (ERS) program which allowed Australians to withdraw up to A$20,000 (roughly 40% of median total income) from private retirement accounts known as superannuation.[2] The ERS program was one of the most significant support measures introduced during the pandemic, and large enough to have a macroeconomic effect. Between April and December 2020, the Australian Taxation Office approved a total of $37.8 billion in withdrawals for 3.05 million people while the program was running, amounting to around 2% of GDP (ATO, 2023a).

Such large withdrawals are likely to have had an impact on people's labour supply decisions, with standard models predicting a contraction in labour supply due to the policy. The large increase in liquidity may have increased workers' reservation wages, leading them to withdraw from the labour market (Borjas, 2016, p. 512). A range of studies have documented the relationship between higher unemployment insurance and longer job search. Chetty (2008) finds this effect to be stronger among liquidity constrained households, arguing that liquidity constraints account for 60% of the increase in unemployment durations caused by more generous unemployment insurance.

Additionally, while the policy did not affect permanent income, the withdrawals could have been treated similarly to an increase in non-labour income, leading people to reduce their labour supply in response (Borjas, 2016, p. 36).[3] Understanding the extent to which this occurred is important to inform whether similar policies should be considered in the future. It is possible that a reduction in the responsiveness of labour supply to wages and job offers stemming from the policy may have contributed to chronic labour shortages which characterised the initial stages of the economic recovery from COVID-19.

A novel contribution of this paper is to identify the effect of the ERS program on labour supply. While previous studies have examined the impact of the ERS program on consumption (Hamilton et al., 2023) and unemployment durations (Sainsbury et al., 2022), we are the first to directly explore its effect on hours worked and employment. Unlike previous studies which used administrative data, we opt for a comprehensive longitudinal panel dataset, the Household, Income and Labour Dynamics in Australia (HILDA) survey. While this data does not have population-wide coverage of policy participants, it records a rich set of information not available in Australian administrative data, such as hours worked and a broad set of demographic and behavioural variables.

We find that the ERS program was associated with a substantial contraction in labour supply for one year after its introduction. We provide causal evidence on the decline in labour supply utilising an instrumental variable (IV) strategy that instruments withdrawal with a measure of impatience. In our baseline specification, our local average treatment effect (LATE) estimates suggest that, for the subset of compliers, the ERS program reduced the probability of employment in 2020 by 33 percentage points, an effect which dissipated in the following year. Additionally, this effect appeared to be concentrated in female withdrawers (reducing employment probability in 2020 by 53 percentage points) and, consistent with our impatience

2 All dollar amounts henceforth are presented in Australian dollars.

3 Assuming leisure is a normal good.

instrument, those who made the full withdrawal amount of $20,000 allowable under the policy (reducing employment probability in 2020 by 47 percentage points).

Our findings are broadly consistent with existing literature on the ERS program. Research into the program's effects on unemployment durations (Sainsbury et al., 2022) finds that the first $10,000 withdrawn was associated with a 32% lower exit rate from unemployment benefits within the first six months of becoming unemployed and a 14% lower exit rate inside of a year. Exit rates between withdrawers and non-withdrawers only reconverged 18 months after the initial withdrawal.

The literature supports our assertion that the ERS program represented a large relaxation in people's liquidity constraints, a mechanism through which we argue it affected labour supply. Surveys of program participants find that three-quarters of withdrawers accessed their superannuation to meet an immediate liquidity need, and few appear to have considered the long-run effect on account balances (Bateman et al., 2023). Characteristics of withdrawers may explain this: they were more likely to be younger, more financially vulnerable, precariously employed, single parents, in poor health and have poor financial literacy (Preston, 2022).

The literature points to many of the people withdrawing being present-biased, or impatient. An interesting feature of the ERS program is that it did not represent an increase in permanent income, but rather provided access to previously illiquid personal savings. While withdrawals may have had the effect of significantly reducing future retirement savings (the Australian Treasury (2020) estimated that a 30 year old withdrawing $20,000 would have on average $69,300 less in retirement in current dollars), a plausible explanation for the high uptake of the program is the existence of present-biased consumers. Indeed, Hamilton et al. (2023) argue that the large consumption response to the policy can only be explained by present bias. They find that withdrawals were associated with a very large spending response in the first four weeks post-withdrawal, indicating that withdrawers had a high marginal propensity to consume.[4]

Our results contribute to a small international literature on the effects of other early pension release schemes. Similar programs have been used to offset economic shocks, such as Chile in response to COVID-19 (Lorca, 2021), Fiji in response to Cyclone Winston in 2016 (Guo and Narita, 2018), and Denmark in response to the global financial crisis of 2008 (Kreiner et al., 2019). However, these studies focused on the effects on retirement balances or consumption responses rather than labour supply. An exception is a 2016 pension reform in Peru that allowed people to withdraw their entitlements as a lump-sum rather than an annuity. While introduced in a non-crisis context, this was found to bring forward retirement due to its alleviation of liquidity constraints (Moreno and Slavov, 2024).

Finally, we also contribute to a broader literature on the impact of large transfers on labour supply. While not exactly resembling the ERS program as transfers increase permanent income, these studies generally find negative effects on labour supply. In particular, several studies have investigated the impact of lottery winnings on labour supply, finding large effects on employment income and hours worked which can persist for a number of years after

4 For a contrasting view, the e61 Institute (2024) found that the per dollar expenditure response of the ERS was smaller relative to the one-off $750 Economic Support Payment and the $550 fortnightly JobSeeker Payment Coronavirus Supplement. Nevertheless, the ERS was found to have a larger aggregate spending effect than either of these support payments, as the average ERS payment was significantly larger.

winning (Imbens et al., 2001; Picchio et al., 2018; Golosov et al., 2024). To the extent that present-biased withdrawers did not represent forward-looking Ricardian agents, our results are consistent with this literature.

This paper is structured as follows. Section 2 provides a discussion of the contextual and institutional setting surrounding superannuation and the ERS program. Section 3 discusses the data and Section 4 details the identification strategy. Section 5 provides the empirical findings and Section 6 concludes.

2 Institutional setting

Australia's retirement system consists of three pillars: a means-tested Age Pension that serves as a 'safety net' close to the poverty line, a compulsory superannuation retirement system, and voluntary savings (Preston, 2022). Under the superannuation system, employers make compulsory payments to funds as a proportion of gross employee wages. This was set at a mandatory minimum of 9.5% from July 2014 to June 2021 which has since risen progressively to 12% as of July 2025 (ATO, 2026). Individuals can make additional contributions to their funds, where before-income tax contributions (e.g., via "salary sacrificing") are subject to a 15% tax while after-income tax contributions are not taxed (ATO, 2024).[5]

These funds include those operated by financial institutions, public sector funds for government employees, funds for workers in a particular industry or corporation, and individual self-managed super funds. Like pension saving schemes in many countries, they are highly illiquid by design and are normally only accessible by workers upon retirement or 65 years of age.[6] Their restricted access and mandatory contribution features are intended to act as a safeguard against present overconsumption and other decision-making biases, thus mitigating the risk of old-age poverty (Wang-Ly and Newell, 2022).

During the COVID-19 pandemic, the Australian Government allowed people to withdraw up to $20,000 from their superannuation accounts, roughly 40% of median total income in 2019-20 (ABS, 2022). $10,000 of this was accessible between 20 April and 30 June 2020 and a further $10,000 in a second application period from 1 July to 31 December 2020 (ATO, 2023a). The ATO approved a total of 4.55 million applications for 3.05 million people, or around 1-in-6 working-age people (Hamilton et al., 2023), which led to the early release of $37.8 billion of super, amounting to around 2% of GDP.

Amounts released under this policy were tax-free and did not have to be included in tax returns, nor did they affect means-tested eligibility for other transfer programs. It should be noted that along with the ERS program, the federal government implemented other reforms and support programs. This included an increase in unemployment benefits (JobSeeker), with the requirements for accessing these payments also relaxed. A wage subsidy program (JobKeeper) was introduced to retain links between employers and employees and restart these jobs when the crisis passed. Many who participated in the ERS likely participated in some of these policies as well.

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5 Additional taxes on before-income tax contributions are applied for high-income earners (earning $250,000 or more per year) or those who contribute over a certain cap (set at $27,500 as of July 2021). After-income tax contributions face a tax once they exceed a cap (set at $110,000 as of July 2021).

6 Aside from the early release that was allowed during the pandemic, individuals are allowed to access their funds earlier only under limited circumstances, such as for those with a terminal illness, a permanent incapacity, or on "compassionate" grounds (e.g., to meet medical expenses or to prevent the foreclosure of a home).

While withdrawals were intended to be conditional on losing hours or employment or being a recipient of unemployment benefits, administrative procedures were fast-tracked to expedite funds access. Eligibility was determined by a simple self-declaration on the ATO's online portal and little activity was undertaken to test whether conditions were met (Bateman et al., 2023). The majority of applications were granted, with the ATO (2023a) reporting that 95.2% of total applications and 96.4% of total dollar amounts requested were approved.[7]

In recognition that the policy may have had a detrimental effect on the accumulation of retirement savings, particularly for younger workers, a re-contributions policy was introduced where people who withdrew under the ERS are now allowed to rebuild their savings (between July 2021 and June 2030) up to the amount withdrawn without paying additional tax (ATO, 2023b). This was introduced via a legislative amendment in *Treasury Laws Amendment (More Flexible Superannuation) Bill 2020* in June 2021 and therefore not known until several months after both withdrawal rounds had taken place.

3 Data

3.1 HILDA survey

We use data from the HILDA survey. HILDA is an extraordinarily rich longitudinal panel survey that tracks 17,000 Australian residents annually and has been running continuously since 2001, which continued through the pandemic. HILDA includes a large set of demographic, labour market, and COVID-19 policy-related variables including specific questions on the ERS program and other economic supports such as JobSeeker unemployment benefits and the JobKeeper wage subsidy. It also collects information on superannuation account balances every four years, with 2018 being the most recent pre-pandemic year in which this information was collected.[8]

Our outcome variables are employment status and hours worked. HILDA uses international and Australian Bureau of Statistics (ABS) conventions in determining labour force status. A respondent is classified as employed if they had a job in a four week reference period prior to the interview (ABS, 2023).[9] HILDA records 'hours per week usually worked in all jobs' for the employed. We code those not in employment as working zero hours.

In the survey data, we observe 14,004 working-age respondents in 2019, with a slight attrition down to 13,625 in 2020 and 13,125 in 2021. Questions about the ERS program were first included in wave 20 (2020) and once more in wave 21 (2021). In wave 20, respondents were asked "Prior to 30 June 2020, did you withdraw superannuation under the COVID-19 scheme for early release of super?", along with the amount withdrawn. This identifies withdrawers in the first ERS round (April to June 2020). In wave 21, the question was reworded to "During the last financial year, did you withdraw superannuation under the COVID-19 scheme for early

7 Amounts approved for release may have differed from the actual amounts released from super funds, e.g., if an individual requested $10,000 and only had $5,000 in their account, then the fund only released $5,000. Moreover, applications were rejected if a person attempted to submit multiple applications in the same financial year and were suspended if there were errors. In some cases, these applications were later approved.

8 See the online appendix for the complete set of HILDA variables used in this paper (Table C.1).

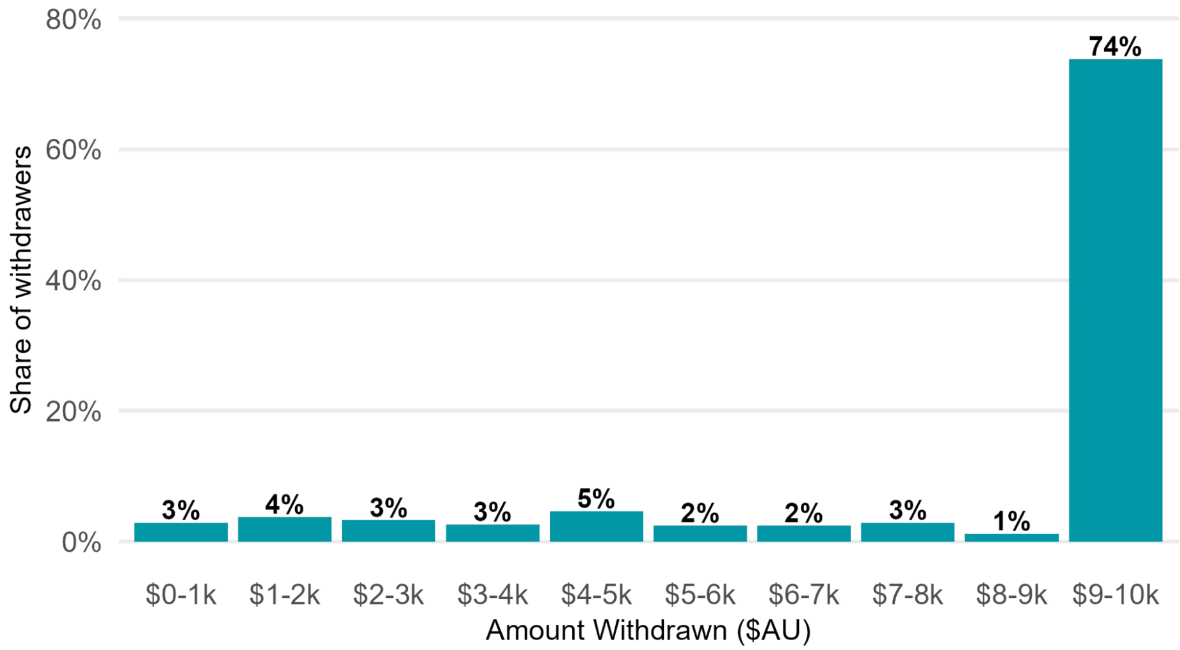
9 A worker is classified as employed part-time if their usual weekly hours of work in all jobs is less than 35 in total. A casual worker is identified as a worker with no paid holiday leave and no paid sick leave.

release of super?", again with the amount withdrawn. Likewise, this identifies withdrawers in the second ERS round (July to December 2020).

Of the working-age population, 10% of HILDA respondents (1,360 individuals) accessed the first round of the ERS program, and 9.6% in the second round (1,264 individuals). This is slightly lower than the 15% share implied by administrative tax data. Of the withdrawers in our sample, 47% withdrew in both rounds, 29% withdrew in the first round only, and 24% withdrew in the second round only.

HILDA also asked withdrawers the amounts they withdrew. Consistent with the administrative data, we see most participants withdrew the full amount of $10,000 in each of the two rounds of withdrawals (Figure 3.1).

Figure 3.1: Superannuation amounts withdrawn (both rounds)



Source: HILDA 2023, own calculations.

<u>3.2 Characteristics of withdrawers and non-withdrawers</u>

We began by inspecting the working-age sample in HILDA before we applied data filters for econometric analysis. We use survey weights in reporting the descriptive statistics in this section.[10] In 2019, prior to the pandemic, 75.5% of our HILDA working-age sample was employed. Conditional on being employed, the average weekly hours worked was 36.4 hours. 21.1% had casual contracts (similar to zero-hour contracts in the UK), and 32.0% were part-time employees.

Some demographic and economic differences can be observed between working-age withdrawers and non-withdrawers (see Online Appendix Table A.1). Withdrawers in our

10 Descriptive statistics in this section use the HILDA survey's "responding person weight", which is the cross-sectional population weight for all people who responded to a particular wave of the survey. See the HILDA User Manual (Summerfield et al., 2023).

sample were slightly less likely to be female (46% compared to 51% of non-withdrawers) and more likely to have dependents (49% compared to 35%).

They were more likely to be employed (85% compared to 74%), in casual employment (22% compared to 15%), and were less likely to have a university degree or higher (22% compared to 32%).

Withdrawers also had a somewhat lower rate of home ownership (50% compared to 58%), as well as noticeably higher debt and lower savings (in 2018). Average weekly wage, hours of work, and years of job experience are comparable across both the sample of withdrawers and non-withdrawers.

We also examined withdrawers and non-withdrawers in HILDA's working-age population sample in terms of their industry and occupation of employment in 2019 (see Online Appendix Table A.2). Withdrawers were slightly more likely to be employed in manufacturing, construction, accommodation and food services, and transport, postal and warehousing, which were industries more affected by lockdowns. They were also far less likely to be professionals, and more likely to be technicians, trade workers and labourers.

4 Empirical strategy

4.1 Sample construction

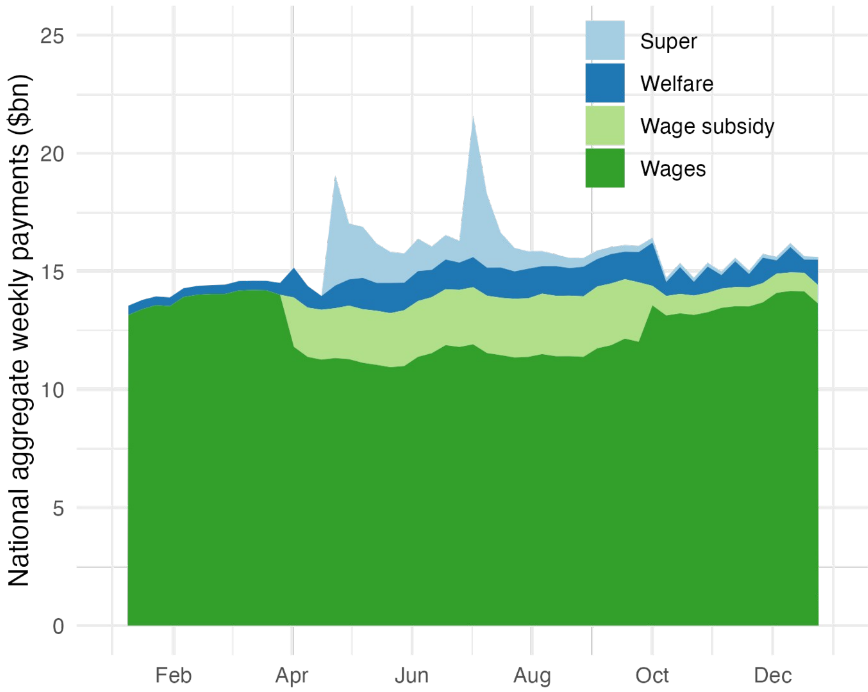
We constructed our sample for analysis from all respondents to the 2019 survey who were of working age (15 to 64 years) and who provided valid responses to their labour market status, age, sex and State/Territory of residence in that year. This resulted in 11,028 unique observations in 2019. Relevant longitudinal data from previous years' surveys, as well as responses from 2020 and later survey waves were appended to this dataset. Some survey attrition was detected, with a 5.9% attrition rate from 2019 to 2020 (a drop of 649 respondents), and a 10.8% attrition rate from 2019 to 2021 (a drop of 1,189 respondents).

4.2 Identification and instrumental variable approach

As with many COVID-19 policies, identification in this setting is challenging. The main threat to identification stems from the fact that people were able to self-select into withdrawing superannuation. This deviates from an ideal experimental setup where random people would have been selected to receive mandatory withdrawals from their superannuation accounts. Reverse causality is an additional concern: people may have withdrawn because they lost hours of work due to the pandemic, or may alternatively have reduced hours because they withdrew.

Our choice of instrument for ERS withdrawals is a measure of impatience. This is motivated by findings from Hamilton et al. (2023) which show that other government supports, in particular JobKeeper and JobSeeker, filled the income gap created during the economic crisis, with ERS withdrawals increasing total liquidity to levels much higher than they had been prior to the pandemic (Figure 4.1). Given incomes remained roughly stable, one plausible explanation for early withdrawals is, therefore, a bias towards present consumption. Our empirical strategy aims to uncover the local average treatment effect (LATE) for those who have withdrawn due to their impatience. However, it is worth noting there may be groups who had other reasons to withdraw, and the LATE will not capture effect sizes for these cohorts.

Figure 4.1: Working-age payments and ERS withdrawals during COVID-19



Source: Hamilton et al. (2023).
Note: Data from Australian Taxation Office and Department of Social Services

We construct our IV with a question in HILDA that asks about people's financial planning behaviours. Our IV is a dummy variable equal to 1 when people report that they plan their savings and spending over the next week, as opposed to longer time horizons.[11] We assign those respondents in our sample who planned more than a week ahead as 'patient' and those who planned only a week ahead as 'impatient'. This construction results in the identification of 7,982 'patient' and 1,994 'impatient' units in our sample.

This financial planning horizon is feasible in the Australian context as weekly and fortnightly pay cycles are common (Fairwork Ombudsman, n.d.). Moreover, Australia has the second highest rate of part-time employment among OECD countries (OECD, 2024), enabling more frequent pay cycles. In our sample, 21.2% of individuals observed in 2019 were classified as impatient.[12] This dummy variable has been applied as a proxy of the rate of time preference in prior research using HILDA (Jetter et al., 2020; Clark and Zhu, 2023).

Our instrument allows us to identify a LATE for a subpopulation of compliers who switch from not withdrawing to withdrawing because of their impatience. The monotonicity condition plausibly holds as it is highly unlikely that 'defiers' exist (i.e., those who would have withdrawn had they been more patient, but decide not to withdraw if they are more impatient). While the information on impatience is only available in 2018, prior to the onset of COVID-19, this lagged

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11 Every two years respondents are asked in an optional self-completion questionnaire (SCQ) component of HILDA "In planning your saving and spending, which of the following time periods is most important to you?" with the options being: (1) "next week", (2) "the next few months", (3) "the next year", (4) "the next 2 to 4 years", (5) "the next 5 to 10 years", and (6) "more than ten years ahead". We construct a dummy variable for impatience from the 2018 data which is a 1 if respondents answered "next week" and a 0 otherwise.

12 We tested an alternative way of classifying impatience by classifying those who also answered "the next few months" as impatient. This alternative did not produce any statistically significant estimates. See Appendix B in the online appendix.

feature of the measure has the attractive quality that it should be less correlated with economic outcomes during the pandemic.

4.2.1 Relevance

We argue patience is relevant to withdrawal as less patient individuals should want to bring forward their consumption. Supporting the relevance of our instrument, 20.9% of individuals in our sample classified as 'impatient' (i.e., those who had a planning horizon of 'next week') withdrew, compared to only 9.5% of those classified as 'patient' (Table 4.1).

Table 4.1: Percentage of withdrawers and non-withdrawers by patience classification, working-age sample

	Patient (n = 7,982)	Impatient (n = 1,994)
Did not withdraw	90.5%	79.1%
Withdrew	9.5%	20.9%
Total	*100%*	*100%*

Source: HILDA 2023, own calculations. Note that sample filters were applied (see Section 4.1).

Moreover, of those who responded to this question in both the 2016 and 2018 HILDA, 82% of those we assigned as patient or impatient did not change their status between these years, suggesting the measure is persistent over time.

4.2.2 Exogeneity

The exogeneity of our instrument requires that impatience is not correlated with the fall in labour supply caused by the withdrawal of superannuation except through the withdrawal of super. It is therefore important to clarify the main channels through which impatience may affect labour supply, and control for these appropriately. This means we can still derive causal LATE estimates, albeit under a *conditional* exogeneity assumption.

One of the strengths of HILDA is that its richness means there is a wide range of potential control variables that can be used to meet the requirements of the conditional exogeneity assumption. There are several potential violations of exogeneity in our context.

- Patient people may be more likely to undertake higher levels of education, as this is generally time-consuming and substitutes immediate income for higher future income. More highly educated people are also more likely to be in more stable forms of employment, meaning the employment of patient people could have been less affected during COVID-19. We address this channel through our controls for occupation and total income, both of which are highly correlated with educational attainment and should absorb most of this variation.

- People could be in a financially precarious position because of their type of employment, in particular being employed on a casual basis. As it is easier for employers to reduce hours of casually employed workers, this means impatience could be directly related to the change in hours caused by COVID-19. In order to mitigate these issues, we control for casual employment and self-employment status.

- Finally, people with an external locus of control (LOC) may believe outcomes are determined by forces outside of their control rather than their own effort. If people with external LOC are more financially short-sighted, this might imply higher ERS uptake. Indeed, Caliendo et al. (2015) and Berger and Haywood (2016) find that external LOC is associated with lower labour market search and lower reservation wages. Similarly, conscientiousness has been found to be a strong predictor of labour market outcomes (Alderotti et al., 2023). If external LOC and conscientiousness have direct labour market effects, then including these as control variables may assist in satisfying conditional exogeneity. However, we include these variables as a robustness check rather than in our baseline specification, as their correlation with impatience might reflect the same underlying personality construct.

More details of control variable construction are contained in the online appendix.

The exogeneity of impatience as an instrument is supported by empirical evidence which indicates time preferences are formed early in life. For example, Bettinger and Slonim (2007) and Andreoni et al. (2019) find that time preferences develop substantially during childhood, with younger children exhibiting greater impatience compared to older children. Moreover, some studies have found time preferences for adults are stable over time, with preferences remaining unchanged over several months and even years (Kirby, 2009; Meier and Sprenger, 2015). Meier and Sprenger (2015) also find that time preferences remain stable despite changes to income, future liquidity, employment and family structure, although this literature is much more mixed (Chuang and Schechter, 2015).

Similarly, the relevance of the instrument is supported by findings that impatience has a strong influence on financial decision-making in a variety of areas. For example, more present-biased individuals are less likely to invest in financial literacy (Meier and Sprenger, 2013) and more likely to have higher credit card debt, even when controlling for disposable income, socio-demographic factors, and credit constraints (Meier and Sprenger, 2010).

Table 4.2 shows some differences in 2019 characteristics between patient and impatient individuals. Supporting the exogeneity assumption, both groups have very similar hours worked in 2019, prior to the influence of any channel of impatience on hours through ERS withdrawals.

Some differences exist between patient and impatient people. We see that patient individuals are more likely to have a partner and children. They are also significantly more likely to be high-skilled, employed and earn a higher mean wage. They also had a higher level of savings and a higher rate of home ownership in 2018. To address any potential threats to instrument validity stemming from these factors, we include these as controls in our IV estimation.

Table 4.2: Attributes of patient and impatient individuals in HILDA, 2019 key variables

	Patient (n = 9,169)		Impatient (n = 2,476)	
Demographic	**Value**	*SE*	**Value**	*SE*
Age (mean)	40.0	*0.183*	39.7	*0.446*
Female (%)	52%	*0.007*	50%	*0.015*
Has partner (%)	61%	*0.007*	46%	*0.015*
Has dependents (%)	41%	*0.007*	34%	*0.014*
Labour force status	**Value**	*SE*	**Value**	*SE*
Employed (%)	81%	*0.006*	65%	*0.014*
Unemployed (%)	3%	*0.002*	6%	*0.005*
Not in labour force (%)	16%	*0.006*	29%	*0.014*
Other labour market indicators	**Value**	*SE*	**Value**	*SE*
Self-employed (%)	10%	*0.004*	7%	*0.008*
Casual worker (%)	15%	*0.005*	18%	*0.011*
Weekly wage (mean)	$1,092	*16.2*	$656	*19.4*
Weekly hours (mean)	36.6	*0.206*	35.5	*0.507*
Years of work experience (mean)	18.2	*0.180*	15.7	*0.348*
Skill level	**Value**	*SE*	**Value**	*SE*
University or higher (%)	38%	*0.007*	15%	*0.012*
Vocational training (%)	31%	*0.006*	37%	*0.015*
High school or less (%)	31%	*0.006*	47%	*0.015*
Other financial metrics (2018)	**Value**	*SE*	**Value**	*SE*
Home ownership in 2018 (%)	62%	*0.007*	44%	*0.015*
Debt in 2018 (mean)	$8,610	*581.1*	$6,116	*603.0*
Savings in 2018 (mean)	$31,269	*1,586.1*	$8,923	*1,152.3*

Source: HILDA 2023, own calculations.
Note: HILDA survey weights (responding person population weight) have been applied. Categories may not add up 100% due to rounding and missing observations. SE: Standard Error.

Our IV equation takes the following form:

$$y_{it} = \alpha + \beta \, withdrew_i + X_i^T \gamma + \varepsilon_i$$

Where α is the constant and ε_i is the individual error term. For individual i we estimate how two different labour market outcomes depend on withdrawing in 2020 ($withdrew_i = 1$ if i withdrew through the ERS and 0 if they did not):

- y_{it} as $employment_{it}$: The probability of employment, coded 1 for i if they are employed and 0 otherwise (unemployed or not in the labour force) at t = 2020 (or 2021 in some specifications). This is estimated as a linear probability model.
- y_{it} as $hours_{it} - hours_{i2019}$: The change in usual weekly hours worked between 2019 and t.

The parameter of interest is therefore β. We include a rich set of other controls (X_i^T) for factors that could plausibly influence the decision to withdraw. These controls include demographic characteristics, state and territory of residence, industry of employment, and labour market status. The complete set of controls are listed below in Table 4.3.

We also run an OLS version of this model as a comparison. As some of our variables (including impatience, LOC and conscientiousness) are sourced from the Self Completion Questionnaire (SCQ), we weight all our models by SCQ responding person population weights.

Table 4.3: Complete list of controls

Controls - baseline specification	Additional notes
Age	
Casual employment	Binary indicator.
Has dependent children	Binary indicator.
Home ownership	Binary indicator. Most recent observation prior to COVID-19 is 2018.
Industry of employment	Industry categories by 1-digit ANZSIC 2006 Divisions.
Labour force status	Broad employment status (employment, unemployment, not in the labour force).
Living with a partner	Binary indicator.
Net savings	Most recent observation prior to COVID-19 is 2018.
Occupation	Occupation categories by 1-digit ANZSCO 2006 Major Groups.
Private income	
Self-employment	Binary indicator.
Sex	Binary indicator.
State/territory of residence	
Superannuation balance	The most recent observation prior to COVID-19 is 2018. HILDA only reports in ranges rather than as a continuous variable.[13]
Total income	Includes wages and non-labour income sources.
Value of all super funds	Most recent observation prior to COVID-19 is 2018.
Controls - robustness check only	**Additional notes**
Conscientiousness score	Time-invariant score per respondent, averaged across respondent scores across multiple survey waves. SCQ variable.
LOC score	Time-invariant score per respondent, averaged across respondent scores across multiple survey waves. SCQ variable.

All these controls are measured in 2019, except for home ownership, superannuation balance, and net savings which are from 2018 as these financial variables are only collected every four years in HILDA. Further details on the source and construction of control variables can be found in the online appendix.

As discussed above, as a robustness check, we also control for external LOC and conscientiousness. Due to the fact some variables are not observed for all units, 8,330 units were used for the IV model with the baseline controls while 7,913 units were used for the IV model with the additional personality controls.

5 Results

Table 5.1 reports the results of the first stage of our instrumental variable regression. The first-stage regression shows that being impatient increases an individual's probability of withdrawing by around 9%, and this estimate is stable across both the baseline and personality

13 These ranges are: ‘Has no super funds’, ‘Less than $5,000’, ‘$5,000 to $19,999’, ‘$20,000 to $49,999’, ‘$50,000 to $99,999’, ‘$100,000 to $199,999’, ‘$200,000 to $499,999’, ‘$500,000 to $999,999’, ‘$1,000,000 to $1,999,999’, and ‘$2,000,000 to $4,999,999.’

control specifications. The correlation supports our claim that our patience instrument is relevant to withdrawal.

Table 5.1: First stage results - effect of impatience on withdrawal

	IV - baseline controls	IV - with personality controls
First stage estimate	0.088*** (0.009)	0.089*** (0.009)
N	8330	7913
Adjusted R2	0.080	0.086
Control variables:		
Baseline	X	X
Conscientiousness & LOC		X

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001
Source: HILDA 2023, own calculations.

5.1 Effect of the ERS on the extensive margin of employment

Table 5.2 presents estimates of the effect of ERS withdrawal on the probability of employment in 2020. The OLS estimates in columns (3) and (4) suggest that withdrawal was associated with a 6-7 percentage point reduction in the probability of employment, and are precisely estimated and stable across both control specifications. The IV estimates in columns (1) and (2) are considerably larger in magnitude, suggesting withdrawal caused a 20-33 percentage point reduction in employment probability. This divergence between OLS and IV estimates is consistent with the interpretation that the instrument identifies a LATE over a sub-group of compliers whose withdrawal decisions are responsive to impatience. As such, their labour supply responses plausibly exceed those of the average withdrawer. We discuss the implications of the instrument tests in Section 5.3.

Table 5.2: Regression of ERS withdrawal on employment probability in 2020

	(1) IV	(2) IV	(3) OLS	(4) OLS
ERS Withdrawal	-0.328** (0.115)	-0.200+ (0.112)	-0.061*** (0.011)	-0.073*** (0.011)
Adjusted R^2	0.184	0.238	0.263	0.279
N observations	8330	7913	8773	8259
First stage F-statistic	90.599	88.665	-	-
Wu-Hausman test p-value	0.0222*	0.279	-	-
Control variables:				
Baseline	X	X	X	X
Conscientiousness & LOC		X		X

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001
Source: HILDA 2023, own calculations.

We find that these results are also robust to changing the definition of working age from 15-64 to 25-54, which trims those who have yet to enter the labour market or are close to the lower survey threshold, as well as those nearing retirement (see Online Appendix D).

Additionally, we rerun our IV approach to estimate the effect of withdrawal on the probability of employment in 2021 to examine whether the negative extensive margin effect has any persistence (see Online Appendix E). We observe no statistically significant effect for 2021, indicating that the policy's reduction in labour supply was a brief shock rather than a persistent effect.

5.2 Employment effects by extent of withdrawal

As a further test of the liquidity mechanism motivating our analysis, we examine whether the employment effects of the ERS are larger among those who withdrew the full amount available. If the reduction in employment probability is driven by the relaxation of liquidity constraints raising reservation wages, we would expect effects to be stronger among those who received the largest transfer. We therefore re-estimate our IV specifications restricting the treatment group to individuals who withdrew the maximum available amount, excluding partial withdrawers from the analysis. We do not present separate results for partial withdrawers given the small number of observations in this group, which would produce unreliable estimates.

Table 5.3 reports the results. The IV estimates are larger in magnitude than those reported in Table 5.2, with the baseline specification suggesting a 47 percentage point reduction in employment probability among full withdrawers, compared to 33 percentage points for the complete sample. This pattern is consistent with a liquidity effect where those who extracted the largest amount from their superannuation experienced the sharpest reduction in labour force participation. We discuss the results of instrument tests reported in Table 5.3 in Section 5.3.

Together, these results provide additional support for the interpretation that the ERS program reduced employment by relaxing liquidity constraints, with effects scaling broadly with the size of the transfer received.

Table 5.3: Regression of ERS withdrawal on employment probability in 2020, full withdrawers only

	(1) IV	(2) IV	(3) OLS	(4) OLS
ERS Withdrawal	-0.467** (0.149)	-0.330* (0.153)	-0.047*** (0.014)	-0.046*** (0.014)
Adjusted R^2	0.155	0.217	0.266	0.281
N observations	7919	7525	8326	7843
First stage F-statistic	84.078	73.799	-	-
Wu-Hausman test p-value	0.00363***	0.0628+	-	-
Control variables:				
Baseline	X	X	X	X
Conscientiousness & LOC		X		X

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Source: HILDA 2023, own calculations.

Similar to withdrawal extent, it would also be attractive to estimate separate models for participants withdrawing in each of the two rounds of the ERS. Unfortunately, the timing of the HILDA data collection cycles makes this infeasible. The first round of ERS withdrawals ran from April to June 2020 and the second from July to December 2020, but the 2020 wave of HILDA was collected from August 2020 through to early 2021, which overlaps with the second round itself. As such, a respondent's recorded labour supply outcomes may capture either a pre- or post-second-round state, depending on their interview date. Moreover, the 2021 wave, collected from July 2021 through to early 2022, is too far removed from either round to recover timing variation.

5.3 Instrument validity

Our instrument is strong across all four of the above IV specifications in Tables 5.2 and 5.3, with first-stage F-statistics well above conventional thresholds for weak instrument concerns (Stock and Yogo, 2005). In our baseline specification, the Wu-Hausman test (Hausman, 1978; Wu, 1973) rejects exogeneity for the complete sample as well as the sample with full withdrawers only. In our specification with personality controls, the Wu-Hausman test only rejects at the 10% level for the sample of full withdrawers, and does not reject for the complete sample. The stronger result for full withdrawers may be due to the instrument more cleanly identifying the effect for full withdrawers, as the most impatient individuals may have been more likely to withdraw the full amount. While this implies that caution should be taken in interpreting the personality control specification for the complete sample, this result may also be due to the smaller effect size for this group combined with the lower precision inherent in the IV estimator.

While the IV results show a noticeable impact on labour participation from the policy, it is worth emphasising that the instrument isolates the LATE for a specific group of compliers who withdrew due to an extremely low planning horizon (i.e., “next week”). A potential reason for the large estimated effect size is that this does not capture the effect on labour supply for ‘never takers’ who, if hypothetically forced to withdraw funds, may have re-contributed the funds immediately, leaving little to no effect on labour supply. It also excludes ‘always takers’ who, regardless of their patience level, would have always participated and may have had a different labour response to the complier group.

Although our IV estimate may lack external validity beyond compliers, it nevertheless provides strong evidence for a negative causal effect of the policy on labour supply. Moreover, the LATE is still relevant from a policy perspective given that a significant proportion of our sample (a fifth) has been identified as impatient. Policies such as the ERS likely did not consider the potential behavioural responses induced by agents with varying innate time preferences. Understanding these diverse behavioural tendencies can enhance compliance with policy conditions and improve overall policy efficiency.

5.4 Extension: Effect of the ERS on the intensive margin of employment

Table 5.4 reports estimates of the effect of ERS withdrawal on weekly hours worked in 2020, restricted to individuals who were employed in 2019. The OLS estimates in columns (3) and (4) suggest withdrawal was associated with a reduction of approximately 4 hours per week, and are precisely estimated and consistent across control specifications. However, the IV estimates in columns (1) and (2) are positive, though statistically insignificant, and the Wu-Hausman test rejects exogeneity at the 5% level in both IV specifications. This indicates that OLS estimates of the intensive margin effect are likely to be biased, and that once the endogeneity of withdrawal is accounted for, we find no statistically significant evidence that the ERS reduced hours among those who remained employed.

Table 5.4: Regression of ERS withdrawal on weekly hours worked in 2020, conditional on being employed in 2019

	(1) IV	(2) IV	(3) OLS	(4) OLS
ERS Withdrawal	6.772 (5.018)	7.319 (5.087)	-3.971*** (0.497)	-3.666*** (0.521)
Adjusted R^2	0.179	0.168	0.269	0.247
N observations	7109	6813	7449	7093
First stage F-statistic	83.613	83.489	-	-
Wu-Hausman test p-value	0.0194*	0.0159*	-	-
Control variables:				
Baseline	X	X	X	X
Conscientiousness & LOC		X		X

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Source: HILDA 2023, own calculations.

Taken together, the extensive and intensive margin results point to a consistent interpretation: the ERS program reduced labour supply primarily by inducing workers to exit employment entirely, rather than by reducing hours among those who remained in work. This is consistent with the theoretical mechanisms motivating our analysis. A large liquidity transfer raises workers' reservation wages, making non-participation preferable to employment at the offered wage. This is a discrete margin of adjustment rather than a marginal one. Present-biased individuals receiving a windfall are likewise more likely to fund a period of non-work than to negotiate incremental hours reductions.

5.5 Extension: Employment effects by sex

Table 5.5 reports IV estimates of the employment effect separately by sex. The results suggest the negative employment effect of ERS withdrawal may have been larger for women than for men, with the baseline IV estimate of -53 percentage points for women being more than twice the magnitude of the corresponding male estimate of -24 percentage points. However, these estimates are only statistically significant at the 5% and 10% levels respectively in the baseline specifications, and neither retains significance when personality controls are added in columns (2) and (4). The reduction in sample size from splitting by sex weakens the precision of these estimates, but the first-stage F-statistic remains high, avoiding concerns that the instrument itself is weaker in these specifications.

Table 5.5: Regression of ERS withdrawal on employment probability by sex

	(1) **IV Female**	**(2)** **IV Female**	**(3)** **IV Male**	**(4)** **IV Male**
ERS Withdrawal	-0.529* (0.224)	-0.331 (0.203)	-0.235+ (0.129)	-0.145 (0.134)
Adjusted R^2	0.112	0.227	0.211	0.242
N observations	4273	4075	4057	3838
First stage F-statistic	29.714	31.793	61.321	54.678
Wu-Hausman test p-value	0.0259*	0.177	0.872	0.258
Control variables:				
Baseline	X	X	X	X
Conscientiousness & LOC		X		X

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Source: HILDA 2023, own calculations.

Notwithstanding these precision limitations, the pattern of results is consistent with existing evidence that women's labour supply decisions are more sensitive to income transfers than men's, reflecting higher labour supply elasticities on average among women (Bargain and Peichl, 2016; Keane and Rogerson, 2015). Women who withdrew may also have been more likely to have caring responsibilities that made a period of non-employment feasible once the liquidity constraint was relaxed. This mechanism is consistent with the higher share of withdrawers with dependents (as noted in Section 3.2). These results should nonetheless be interpreted with caution given the loss of significance under the more conservative specification.

6 Conclusion

This paper has examined whether the early release of pension savings during the COVID-19 pandemic led to a reduction in labour supply due to the large increase in liquid wealth provided by the policy. Using an IV approach, we find that withdrawals reduced labour supply, with the effect being large in 2020 before subsiding in 2021. Our evidence aligns with past research on the ERS program, which found it increased unemployment durations for up to 18 months post-withdrawal (Sainsbury et al., 2022).

Understanding the effects of the ERS program on labour supply has important implications for the design of future economic support packages. Policymakers must keep in mind the potentially large withdrawals of labour supply that could stem from large cash transfers or pension withdrawals. While the spending response of such policies can stimulate economic activity in the short-term (Hamilton et al., 2023), a contraction in labour supply may serve to decrease productive capacity. During the post-pandemic context of global supply shortages, this may have served to prolong the economic recovery.

Our results also contribute to our understanding of pension withdrawals on labour supply more broadly. They suggest that allowing people lump-sum withdrawals may decrease labour supply by more than if such amounts had been paid out as annuities, aligning with existing evidence (Moreno and Slavov, 2024). Such lump-sum withdrawals are currently allowed in Australia upon reaching retirement age as an alternative to a stream of regular payments, which can enable people to spend their lump-sum and then claim the Age Pension. This suggests room for policy redesign if a goal of the Australian retirement system is to promote labour market participation among older workers or to disincentivise uptake of the Age Pension where private savings are sufficient to fund retirement.

Of course, it is also important to recognise this policy was implemented in response to an unprecedented global pandemic event. As such, while hindsight has enabled us to evaluate its labour supply effects, the ERS should still be understood within the context of a suite of relief efforts that Australian policymakers were rapidly implementing during a period of extreme economic uncertainty. Furthermore, while this paper is focused on labour supply, it has not considered the welfare implications of the policy. Increased liquidity may have improved the wellbeing of some who chose to withdraw funds and then reduce their labour supply, if they had a preference for higher consumption in the present over future consumption. Given this possibility, future research is needed to understand both the longer-run ramifications of lump-sum pension withdrawals on labour market engagement and on the welfare outcomes of individuals with differing intertemporal preferences.

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